

**SWINBURNE VIETNAM
ALLIANCE PROGRAM**



Assignment 1:
Research into AI Ethics & Responsible AI

Title: SHOULD AI SYSTEMS BE ALLOWED TO KILL?

Course Name: COS30019 - Introduction to Artificial Intelligence

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I. Summary

This report examines whether artificial intelligence (AI) systems should or should be permitted to use lethal force. These systems are often known as lethal autonomous weapons (LAWS) - that is, weapons once activated, can independently select targets and attack without more human intervention. Key discussed issues include: Whether LAWS accept to follow International Humanitarian Law (IHL), especially principles such as the distinction between civilians and military targets, balance, and preventative solutions; Ethical questions about believing in life-or-death decisions for machines; Strategic risks, such as the potential for conflict growth or the spread of these weapon technologies; and a management structure that could help control or limit high-risk AI systems.

Currently, in severe policy debates, many countries and organizations emphasize the need for “meaningful human control” or “appropriate human judgment” when using life-threatening force. However, there is still much debate about whether this is a legally and ethically mandated requirement or merely a guideline for design and operation within permissible limits.

On balance, the report argues that AI systems should not be permitted to autonomously decide to kill humans, and any use of autonomy in weapons must be strictly bounded by international rules that prohibit autonomous targeting of people and ban unpredictable systems, while restricting other forms to narrow contexts with tight human supervision, accountability, and risk controls. This aligns with recent UN/ICRC calls for new binding rules by 2026, Australia’s push for mandatory guardrails in high-risk AI, and the EU’s AI Act approach to prohibited practices and high-risk uses.

II. Introduction

The rapid progress of AI - perception, planning, and learning - enables weapons that select and strike targets at machine speed. While autonomy already exists in munitions (e.g., fire-and-forget rockets and flying munitions), giving lethal decisions to AI leads to ethical alarm bells and legal uncertainty. As Burton et al. (D1) note, educators and practitioners must grapple with the moral, ethical, and philosophical impacts of AI, including how autonomous decision-makers affect society and human responsibility.

In Australia, the 2024 “Introducing mandatory guardrails for AI in high-risk settings” proposal paper (D2) sets out a risk-based approach, with ten guardrails across the AI life-cycle and regulatory options ranging from sector integration to a new cross-economy AI Act - a framework that closely tracks international movement toward binding limits on high-risk AI.

Key questions addressed in this report are: What are LAWS and where do they sit on the autonomy spectrum? Can LAWS comply with IHL (distinction, proportionality, precautions)? What counts as meaningful human control / appropriate human judgment? What are the strategic risks (escalation, arms race, proliferation to non-state actors)? How do we assign accountability when harm occurs? What can Australia’s protection solutions, EU AI Act, and NIST AI RMF contribute to governance? And what do case studies show about current capabilities and risks?

Methodology and scope.

This research synthesizes peer-reviewed articles, policy frameworks, and international law sources (UN, ICRC, US DoD, EU, UK, NIST, Australia). It focuses on international armed conflict contexts while drawing general governance lessons for high-risk AI beyond the military.

III. Research Findings

3.1. Defining Lethal Autonomous Weapons and the Spectrum of Autonomy

There is no single definition of lethal autonomous weapons (LAWS), but the common point is that these are weapon systems that, once activated, can independently select and attack targets without human intervention. The First Committee of the UN General Assembly and a large number of official UN reports have concerned about allowing algorithms to decide life and death, and have called for states to engage in discussions and establish a common control framework.

Classification of Autonomy Levels:

In policy discussions and analyses, the autonomy of weapons is often divided into three groups:

- Human-in-the-loop (semi-autonomous): The system only suggests targets, and humans must review and approve before firing.
- Human-on-the-loop (supervised autonomous): The system can automatically participate in the attack process, and humans directly monitor and have the right to break or override the system's actions.
- Human-out-of-the-loop (completely autonomous): The system can attack without human permission or monitoring.

This classification is widely used in legal-policy analyses and continues to be repeated in academic reviews in 2025.

The US Department of Defense Directive 3000.09 (2023) does not completely ban autonomous weapons, but gives very strict requirements, including appropriate human judgment when using lethal force, risky testing, high-level evaluation, and following the Department of Defense's AI Ethics Principles and responsible AI development strategy. These requirements aim to ensure that autonomous weapons, if developed, will still operate within a safe, clear, and controllable framework.

3.2. What International Humanitarian Law (IHL) Requires

International Humanitarian Law (IHL) is a list of rules adjusting how warring parties behave in conflict, emphasizing three core principles: the distinction between soldiers and civilians, the proportionality of military action, and preventative solutions to minimize unintended harm. According to the International Committee of the Red Cross (ICRC), allowing automated systems to target people raises serious ethical concerns and makes following the IHL difficult to ensure. For this reason, the ICRC calls for legal rules prohibiting automated weapons with unpredictable behavior and prohibiting systems designed or operated to use force against people. For other cases, the

organization suggests strict limitations on the scope of use, such as location, time, type of target, and level of human monitoring.

The 2023 joint call by the United Nations and the ICRC emphasized a similar position: states need to completely ban automated targeting of humans, as well as AI systems with unpredictable behavior, such as unexplainable machine learning models. From this perspective, ensuring that humans keep control in life-or-death decisions is a hopeless ethical boundary.

3.3. The “Meaningful Human Control” / “Appropriate Human Judgment” Debate

A general agreement has not formed on definitions, but policy exists:

- The ICRC and many states agree with the maintenance of human control over lethal force and specific prohibitions/restrictions.
- The US DoD standard is “appropriate levels of human judgment” throughout the system life-cycle and employment.
- The UK Defense AI Strategy (2022) commits to ambitious, safe, and responsible AI and refuses systems without meaningful, context-appropriate human involvement - even though critics note uncertainty in assessing “context-appropriate.”

At the UN CCW Group of Governmental Experts (GGE), states are working toward elements of an instrument; progress is real but slow under agreed rules, pushing some momentum to the UN General Assembly, which approved its first intention on autonomous weapons in Dec 2024.

3.4. Strategic, Security, and Escalation Risks

Autonomous weapons can decrease the use of force, speed growth, and spread rapidly, including to non-state actors. The UN and ICRC warn that by reducing risk to users' consciousness, LAWS may make conflict more likely and raise arms races.

The US Congressional Research Service (2025) highlights how LAWS could enable operations in limited environments regarding communications but bring legal, ethical, and strategic challenges. Thus, it is essential for global efforts to limit or ban certain systems.

3.5. Accountability, Responsibility, and the Alleged “Gap”

A classic worry is the “responsibility gap” - if an autonomous system kills unlawfully, who is responsible? Philosophers debate whether gaps are inevitable or whether command responsibility and control over probable outcomes are enough for taking responsibility. Some argue LAWS reduce the ethical foundation of Just War by decreasing humans' impact; others argue responsibility can still be traced up the chain of command (procurement, deployment, ROE).

Policy practice is already moving to prevent gaps: DoD 3000.09 emphasizes human judgment, weapon reviews, and careful verification; CCW processes discuss legal reviews and risk reduction; and Australia's rule concepts would tie developers and deployers to life-cycle duty which clarifies accountability across the supply chain.

3.6. Australian “Mandatory Guardrails,” EU AI Act, and NIST AI RMF: Lessons for High- Risk AI

Relevant Policy and Legal Frameworks:

Australia (D2): The Australian suggestion lists out 10 mandatory protections, including risk identification, data administration, testing and security assurance, matter response, production monitoring, etc. In addition, the plan outlines three legal selections: combination into sector law, the creation of a framework law, or the passing of a new AI Act that is applicable across the entire economy. The main key to this is on high-risk AI use cases, and it is suitable with Australia's Voluntary AI Safety Standard as well as human rights contributions from organizations such as the Australian Human Rights Commission.

European Union AI Act (EU AI Act – 2024/2025): The EU AI Act clearly defines prohibited behaviors, such as behavior control techniques or certain forms of biometric data use. For high-risk cases, the Act introduces strict control requirements. Although this isn't a document specifically for the defense sector, the thinking behind defining unacceptable behavior, along with the European Commission's 2025 guidance on unacceptable risks, has created a clear legal framework. This is particularly useful when establishing limits on military autonomy or systems that can be used for both civilian and military purposes (dual-use).

NIST's AI Risk Management Framework (2023): The NIST AI Risk Management Framework 1.0 gives out four key steps: Administration, Mapping, Measurement, and Risk Management. This framework guides organizations in identifying and reducing AI-related problems - including safety, security, and explainability issues. These

factors are directly related to assessing the safety of weapons and whether high-risk AI systems are accepted.

Overall, these policy frameworks support a "defense thinking" method: prohibiting AI uses that break human rights or international humanitarian law, and strictly controlling high-risk applications through solutions such as security assurance, peer testing, issue response, and operator training. These guidelines help create a safer foundation for AI applications in many fields, especially systems with a high degree of autonomy or the potential to cause significant harm.

3.7. Case Studies and Current Systems

Loitering munitions (IAI Harpy/Harop family).

IAI describes Harpy as a fully autonomous, anti-radiation loitering munition that can search for and strike emitting radar targets after pre-programming—often cited as an example of deployed autonomy in weapons. Its newer variants combine anti-radiation homing with EO/IR seekers and sometimes man-in-the-loop control; these illustrate the faded line between autonomy and operator monitor in reality.

Open sources (e.g., US Army ODIN and NGO trackers) likewise describe Harpy as “fire-and-forget,” autonomous detection and attack of emitters, operational with multiple states—showing how narrow target profiles (radars) are more readily automated than human identification.

UN & CCW process momentum (2024–2025).

States at the CCW GGE worked through rolling texts toward elements of an instrument, while the UNGA approved a significant resolution in 2024, encouraging responses to humanitarian, legal, security, and ethical concerns. Australia contributed papers on weapon legal reviews for autonomy—evidence of state reality building toward rule-making.

Policy baselines (US, UK).

The US DoD updated 3000.09 in 2023, making human judgment and assurance requirements legal; the UK (2022) planned for responsible AI in defense, while expressing dislike of no-human-involved systems, although without a precise level, providing a fragmented administration landscape.

3.8. Ethical Analysis: Human Dignity, Just War, and Risk Redistribution

Human and moral agency.

The UN/ICRC's viewpoint is obvious about autonomous targeting of humans: it passed a moral line because machines lack moral mode, cannot apply kindness, and risk inhumane killing. This is the same as parts of Just War theory (*jus in bello*) that prioritize discrimination and proportionality with moral judgment.

Just War debates are complicated.

Some philosophers argue that LAWS could, in principle, meet fairness in risk distribution - for example, reducing risk to soldiers without increasing risk to civilians - if and only if design and rule ensure compliance and responsibility; others debate that responsibility holes make such compliance to be untruthful ethically. The literature remains contested but increasingly accepts the legal difficulties to ensure meaningful human control and maintain responsibility.

Risk management: from ethics to technique.

Practical administration should see autonomous targeting as a critical space for safety: apply ban + restrict. It means that ban AI deciding to use lethal force on persons and unpredictable systems; restrict other autonomous functions to narrow, predictable targets (e.g., fixed military objects like emitting radars), geographical limitation, short time frame, positive operator control, abort/override features, pre-deployment system hazard analyses, model evaluation, and post-deployment monitoring. This illustrates the structure suggested by ICRC/UN, Australia's protection solutions, EU AI Act's prohibited/high-risk split, and NIST AI RMF engineering practices.

IV. Conclusion and Recommendations

Conclusion

The world is gathering on a clear aim: no autonomous killing of humans; no unpredictable machine-learning weapons; and strict limitations on any remaining autonomous weapon functions. Existing IHL duties do not depend on technology but are increasingly hard to meet as autonomy levels increase; therefore, new rules, as encouraged by the UN Secretary-General and ICRC for conclusion by 2026, are essential. Lawful and ethical use requires retained human control, responsibility, and auditability.

Recommendations.

- Approve a Ban-and-Restrict Treaty: Prohibit autonomous targeting of persons and unpredictable ML-based weapons; restrict other autonomous functions with meaningful human control, time/place limits, target-type limits, and real-time monitor/override.
- Create Lifecycle Challenges (Australia's approach): Implement ten protections through legislation or a cross-economy Act; create a registration assessment for high-risk AI, with tasks for both developers and deployers.
- Match with EU AI Act Logic: Use reality-prohibited lists for defense-related uses when suitable; ensure high-risk controls (risk management, data administration, testing, monitoring, incident reporting).
- Establish Safety Engineering: Require DoD-style assurance (V&V, OT&E, failsafes, logs, explainability) and NIST AI RMF practices; mandatory weapon legal reviews for autonomy (Article 36-style) before giving to use.
- Responsibility Chain: Arrange duty of care across the supply chain (developers, leaders); retain command responsibility and ensure investigation and mechanisms for incidents
- Confidence-Building at the UN: Support CCW GGE negotiations while taking advantage of UNGA momentum; commit to clear measures and national policies

Thus, AI systems should not be allowed to make autonomous decisions to kill. Any autonomy in weapons must be followed by meaningful human control with strict rules, national limitations, and verifiable safety cases.

V. Acknowledgements and Resources

- D1 (Burton et al., 2017) helped frame the ethical landscape and the need to integrate ethics into AI design and education, shaping the introductory framing.
- D2 (Australia's 2024 Proposals) informed the guardrails and risk-based regulatory approach, including options for registration and conformity.
- UN/ICRC, CCW documents, US DoD, EU Commission guidelines, NIST AI RMF, UK Defense AI Strategy, and technical/NGO sources provided current policy, law, and engineering baselines.

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